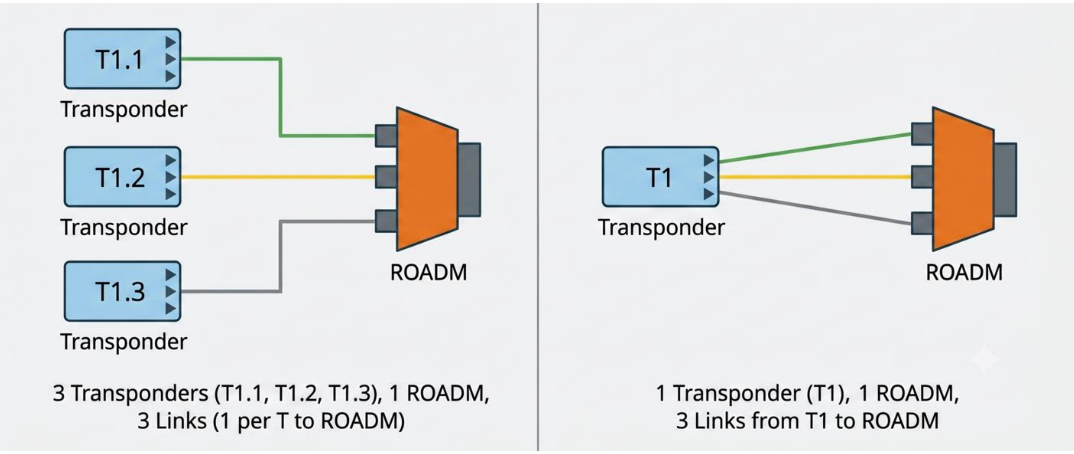
TFS OpticalController

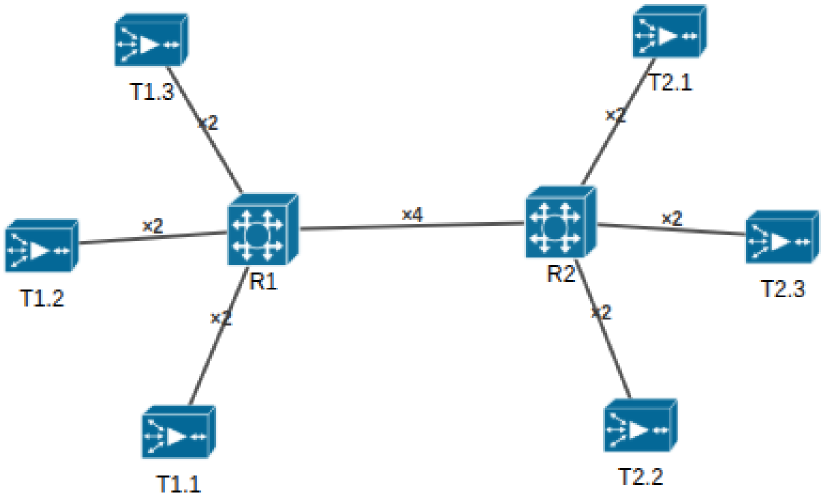
OFFICIAL CONTRIBUTION PROPOSAL

BY: ASM CHAFIULLAH (CHAFI)
SUPERVISED BY: PROF. ALESSIO & PROF ANDREA

JSON Descriptor [Device] & Web View

- Current Format: "T1.1", Proposed Format: "T1"
  (Previous graph generation method had issues with modified device names in the links, ex: R1.5201)
- Topology View: device-port pattern, parallel link representation





Graph Generation [RSA.py]

- Introducing "NetworkX"
- New graph generation method "initGraphNetworkX()"

```
class RSA():
    def __init__(self, nodes, links):
        self.nodes_dict = nodes
        self.links_dict = links
        self.g = None

        self.flow_id = 0
        self.opt_band_id = 0
        self.db_flows = {}
        self.initGraph2()
        self.c_slot_number = 0
        self.l_slot_number = 0
        self.s_slot_number = 0
        self.optical_bands = {}
```

```
class RSA():
    def __init__(self, nodes, links):
        self.nodes_dict = nodes
        self.links_dict = links
        self.g = None

        self.flow_id = 0
        self.opt_band_id = 0
        self.db_flows = {}
        self.initGraphNetworkX()
        self.c_slot_number = 0
        self.l_slot_number = 0
        self.s_slot_number = 0
        self.optical_bands = {}
```

How the graph will be generated?

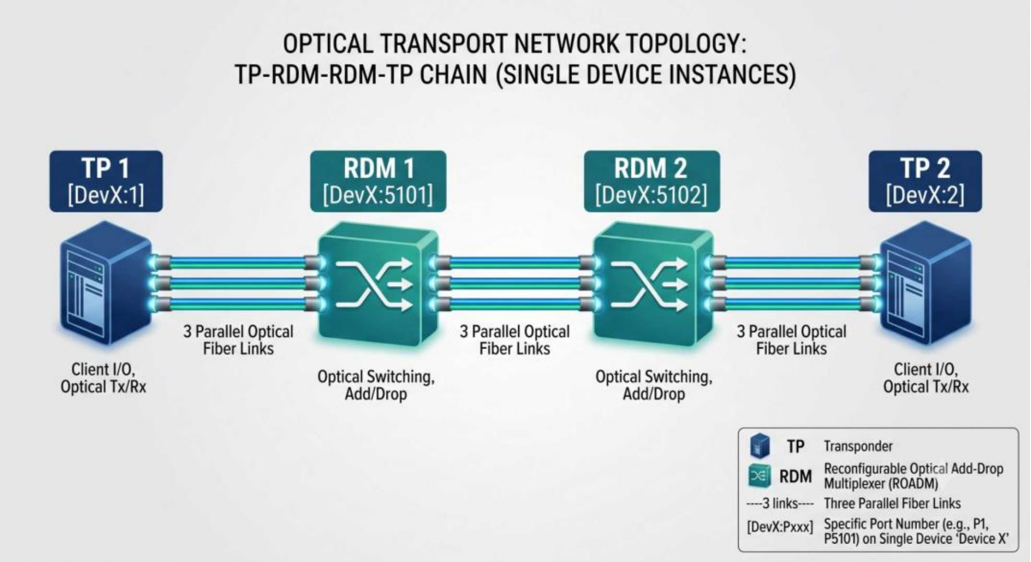
- Singel device with multiple ports

Q: what happens with multiple devices? How the parallel links between TP-RDM will be checked?
[Parallelism in the add/drop links will be checked from the RODM point of view]

- Port based configuration [nodes will connect using device:port configuration]
- Parallel links between devices
- Pre-computation to remove used "add", "drop" links

[This is possible only if we let the controller choose which port to use. If the ports are defined and they are not free, we can refuse the connection.]

- Modification:
  - RSA.py
  - new initGraphNetworkX() method



Path Computation

- Links will be treated only in one direction: unidirectional, bidirectional
- Bidirectionality will be assessed based on service-request.
  (Opposite direction links in Uni-directional configuration will be updated through peer-ports, codes will be re-used)
- Parallel links are individual vertices in networkX [RDM-RDM connections]
- Preference based path selection scheme [Yes/No, Proposed: spectrum-check(SC) scheme]
- Modification:
  – RSA.py
  – method name: compute_path_networkX()

Question:

- Are we computing single path or one extra path for failure management? [can compute a list as well, then run a loop for spectrum check. This is applicable for other schemes except SC.]

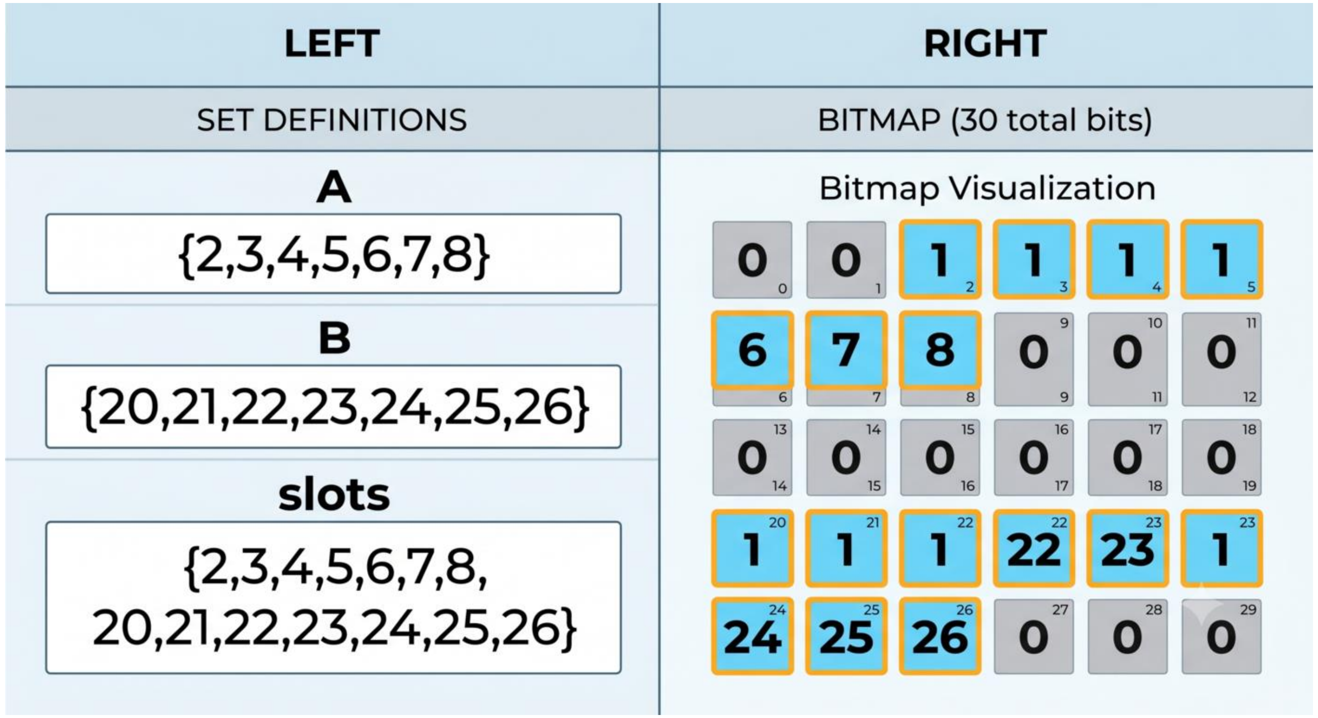
Spectrum Assignment: Frequency Expansion

Question:

- Full band expansion or based on "optical_details: c_slots, l_slots" configuration
- Bandwidth Calculation, Modulation, Operational Mode [right now hard-coded]
- Band choice [currently C is preferred, if preference is not provided]

Spectrum Assignment: Continuity & Contiguity

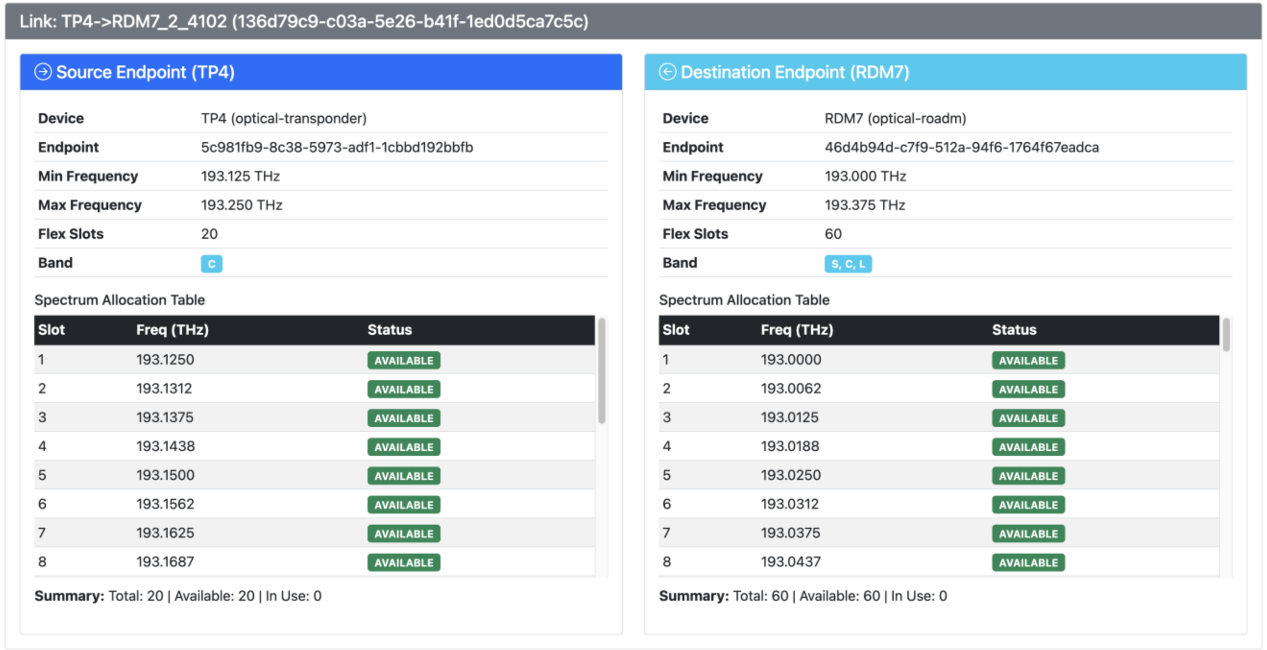
- Offset based set intersection → bitmap-based slot checking
- Contiguity will be checked at the acquisition level (currently checked at link update phase).
- Modification:
  - RSA.py
  - method "select_slots_and_ports2()"



gRPC & NETCONF

- The same configuration will remain

Web View: if approved 🙂


Link: TP4->RDM7_2_4102 (136d79c9-c03a-5e26-b41f-1ed0d5ca7c5c)

Source Endpoint (TP4)

Device	TP4 (optical-transponder)
Endpoint	5c981fb9-8c38-5973-adf1-1cbbd192bbfb
Min Frequency	193.125 THz
Max Frequency	193.250 THz
Flex Slots	20
Band	C

Spectrum Allocation Table

Slot	Freq (THz)	Status
1	193.1250	AVAILABLE
2	193.1312	AVAILABLE
3	193.1375	AVAILABLE
4	193.1438	AVAILABLE
5	193.1500	AVAILABLE
6	193.1562	AVAILABLE
7	193.1625	AVAILABLE
8	193.1687	AVAILABLE

Summary: Total: 20 | Available: 20 | In Use: 0

Destination Endpoint (RDM7)

Device	RDM7 (optical-roadm)
Endpoint	46d4b94d-c7f9-512a-94f6-1764f67eadca
Min Frequency	193.000 THz
Max Frequency	193.375 THz
Flex Slots	60
Band	S, C, L

Spectrum Allocation Table

Slot	Freq (THz)	Status
1	193.0000	AVAILABLE
2	193.0062	AVAILABLE
3	193.0125	AVAILABLE
4	193.0188	AVAILABLE
5	193.0250	AVAILABLE
6	193.0312	AVAILABLE
7	193.0375	AVAILABLE
8	193.0437	AVAILABLE

Summary: Total: 60 | Available: 60 | In Use: 0

- Changes:
  - dropdown based link inspection [why: can be explained, if necessary.]
  - decoration of the band slots [changes made on thesis]
  - lightpath visualization with devices and flow directions
- Modification:
  - only at the web-view optical-view level [no changes in drivers]

Questions?